MSc. Project Report: Survey on Equations over Finite Fields

TIFR 2026

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Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	**Meaning**
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^{\infty} e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^{\top}C_fX$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Decoding from Character Evaluations

Character sums of polynomials over finite fields carry deep arithmetic information, and one of the most striking manifestations of this is that they give rise to good error-correcting codes. The mechanism is simple: two polynomials whose character evaluations are close must be nearly indistinguishable by the character, and the Weil bounds make this precise. For a multiplicative character ψ of $\mathbb{F}_q$ and two polynomials $f, g \in \mathbb{F}_q[X]$ of small degree, the Weil bound controls gives a lower bound on the Hamming distance between the codewords $(\psi(f(\alpha)))_{\alpha \in \mathbb{F}_q}$ and $(\psi(g(\alpha)))_{\alpha \in \mathbb{F}_q}$. The resulting codes turn out to have distance approaching $q/2$ while the message length grows polynomially in q.

Two families are of particular interest. For q odd, applying the quadratic residue character $\psi(\beta) = \beta^{(q-1)/2}$ to squarefree monic polynomials $g(X)$ of degree $d = o(\sqrt{q})$ defines a nonlinear code over $\{0, \pm 1\}$ of length q. For $q = 2^b$, taking $g(X)$ with only odd-degree monomials and forming $\mathrm{Tr} \circ g : \mathbb{F}_q \to \mathbb{F}_2$ gives the classical *dual BCH codes*, introduced independently by Bose–Ray-Chaudhuri [BRC60] and Hocquenghem [Hoc59]. These codes are linear over $\mathbb{F}_2$, have length q and dimension $\frac{d+1}{2} \log_2 q$, and their distance analysis relies entirely on the Weil bound for additive character sums. Dual BCH codes are remarkable: for small d they are *unmatched* codes in the sense that, without the Weil bounds, there is no known proof that binary codes of their parameters even exist.

Despite this theoretical richness, decoding these codes from a constant fraction of errors was a long-standing open problem. The essential difficulty is that each evaluation $\psi(g(\alpha))$ carries very little information about g — the quadratic character discards the magnitude and retains only whether $g(\alpha)$ is a quadratic residue — so the system of equations one must solve is both underdetermined and nonlinear. Even the zero-error interpolation problem, of recovering g exactly from the values $\psi(g(\alpha))$ for all $\alpha \in \mathbb{F}_q$, had no polynomial-time solution for non-constant d prior to the work presented here.

The goal of this chapter is to present the first polynomial-time algorithms for decoding both code families from a constant fraction of errors, following Kopparty. The algorithms combine two classical ideas: the *Berlekamp–Welch* framework, which uses a high-multiplicity error-locator polynomial to zero out error positions and reduces decoding to a linear system, and the *Stepanov polynomial method*, the elementary approach to the Weil bounds that works by interpolating auxiliary polynomials vanishing with high multiplicity at points of interest. The fusion of these two ideas gives rise to what we call *pseudopolynomials*: high-degree polynomials all of whose Hasse derivatives, when evaluated on $\mathbb{F}_q$, agree with polynomials of much smaller degree. These objects turn out to be the correct algebraic framework for capturing the special structure of character evaluations of polynomials.

§ 1.1 Theory of Pseudopolynomials

1.1.1 Pseudopolynomials

Definition 1.1.1: Pseudoderivative

Let $F(X) \in \mathbb{F}_q[X]$. Define the ℓ-th pseudoderivative *of F, denoted $F_{\langle \ell \rangle}(X)$, to be the unique polynomial of degree at most $q-1$ whose evaluations on $\mathbb{F}_q$ agree with the evaluations of $F^{(\ell)}(X)$. Explicitly:*

$$F_{\langle \ell \rangle}(X) = \left(F^{(\ell)}(X) \bmod \Lambda(X)\right),$$

where $\Lambda(X) = X^q - X$.

Abusing notation, we will sometimes use $F_{\langle \ell \rangle}$ to denote the function $F_{\langle \ell \rangle} : \mathbb{F}_q \to \mathbb{F}_q$, defined by $F_{\langle \ell \rangle}(\alpha) = F^{(\ell)}(\alpha)$.

Definition 1.1.2: Pseudodegree

Let $F(X) \in \mathbb{F}_q[X]$. We define the pseudodegree *of F, denoted $\mathrm{pdeg}(F)$, to be:*

$$\max_{\ell \geq 0} \deg(F_{\langle \ell \rangle}).$$

*We say $F(X)$ is a k-*pseudopolynomial *if $\mathrm{pdeg}(F) \leq k$. When $\deg(F) = d$ we refer to F as a k-*pseudopolynomial of degree *d.*

By definition, we have same basic observations on how pdeg behaves on addition and multiplication of polynomials.

Observation 1.1. *Like normal degree of polynomials* pdeg *behave the similar way, but we have inequality instead of equality. For any $F, G \in \mathbb{F}_q[X]$*

- $\mathrm{pdeg}(F+G) \leq \max(\mathrm{pdeg}(F), \mathrm{pdeg}(G))$.
- $\mathrm{pdeg}(F \cdot G) \leq \mathrm{pdeg}(F) + \mathrm{pdeg}(G)$.

1.1.1.1 Algebraic Characterization

Lemma 1.1.1 Algebraic Characterization of Pseudodegree

Let $F(X) \in \mathbb{F}_q[X]$, and let $F(X) = \sum_{i \geq 0} F_i(X)\,\Lambda(X)^i$, with each $\deg(F_i) < q$, be the base-$\Lambda(X)$ expansion of $F(X)$. Then $\mathrm{pdeg}(F) = \max_{i \geq 0} \deg(F_i)$.

Proof: We expand $F^{(\ell)}$ by applying the product rule for Hasse derivatives to each term of the base-Λ expansion:

$$F^{(\ell)}(X) = \sum_{i \geq 0} \left(F_i \cdot \Lambda^i\right)^{(\ell)}(X) = \sum_{i \geq 0} \left(\sum_{\substack{\ell_0, \ell_1, \ldots, \ell_i \\ \sum \ell_j = \ell}} F_i^{(\ell_0)}(X) \cdot \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X) \right).$$

When we reduce mod $\Lambda(X)$, recall that $\Lambda^{(0)} = \Lambda \equiv 0$, and $\Lambda^{(j)} = 0$ for $j \notin \{0, 1, q\}$. So the only terms surviving mod Λ

are those with $\ell_1, \ldots, \ell_i \in \{1, q\}$, which forces $\ell \geq i$ and $\ell_0 = \ell - (\ell_1 + \cdots + \ell_i) \in [0, \ell - i]$. Collecting these terms:

$$F^{(\ell)}(X) \bmod \Lambda(X) = \sum_{i=0}^{\ell} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot \underbrace{\left(\sum_{\substack{\ell_1,\ldots,\ell_i \in \{1,q\} \\ \ell_1+\cdots+\ell_i = \ell-\ell_0}} \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X) \right)}_{=: C_{i,\ell,\ell_0,q} \in \mathbb{Z}}$$

$$= (-1)^{\ell} F_\ell(X) + \sum_{i=0}^{\ell-1} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot C_{i,\ell,\ell_0,q},$$

where the leading term isolates $i = \ell$, $\ell_0 = 0$, the unique case with $\ell_1 = \cdots = \ell_i = 1$, which contributes the factor $(-1)^{\ell}$.

Forward direction. When $\deg(F_i) \leq k$ for all i, each $F_i^{(\ell_0)}$ has degree at most k, so the entire expression $F^{(\ell)} \bmod \Lambda(X) = F_{\langle \ell \rangle}(X)$ has degree at most k. Since ℓ was arbitrary, $\operatorname{pdeg}(F) \leq k$.

Reverse direction. Rearranging the identity above gives

$$(-1)^{\ell} F_\ell(X) = F_{\langle \ell \rangle}(X) - \sum_{i=0}^{\ell-1} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot C_{i,\ell,\ell_0,q}.$$

We argue by induction on ℓ. The base case $\ell = 0$ gives $F_0 = F_{\langle 0 \rangle}$ directly, so $\deg(F_0) \leq \operatorname{pdeg}(F)$. In the inductive step, assuming $\deg(F_i) \leq \operatorname{pdeg}(F)$ for all $i < \ell$, every term $F_i^{(\ell_0)}$ on the right has degree at most $\operatorname{pdeg}(F)$, so $\deg(F_\ell) \leq \operatorname{pdeg}(F)$ as well. Hence, $\max_{i \geq 0} \deg(F_i) \leq \operatorname{pdeg}(F)$. The two directions together give $\operatorname{pdeg}(F) = \max_{i \geq 0} \deg(F_i)$. ■

1.1.1.2 Multiplicities

Now the pseudopolynomials are special because even though we may have a very large degree pseudopolynomials the multiplicities of an irreducible factor is small. The following lemma gives a bound on the multiplicity modulo q.

Lemma 1.1.2 Multiplicity Bound for Pseudopolynomials

Let $k < q$, and let $F(X)$ be a nonzero k-pseudopolynomial of degree D over $\mathbb{F}_q$. For an irreducible polynomial $H(X) \in \mathbb{F}_q[X]$, let μ be the highest power of $H(X)$ that divides $F(X)$. Then:

$$\mu \bmod q \in \left[0,\ k + \left\lfloor \frac{D}{q} \right\rfloor\right].$$

Proof: Pick a root $\alpha \in \overline{\mathbb{F}_q}$ of $H(X)$, so that $\mu = \operatorname{mult}(F, \alpha)$ and $F^{(\mu)}(\alpha) \neq 0$. By Lemma 1.1.1, the base-Λ expansion of F has the form $F(X) = \sum_{i=0}^{t-1} F_i(X)\Lambda(X)^i$ with $t = \lfloor D/q \rfloor + 1$ and $\deg(F_i) \leq k$.

It suffices to show $F^{(\ell)}(X) = 0$ for every ℓ with $\ell \bmod q \in [k + t, q - 1]$, since this would confine $\mu \bmod q$ to $[0, k + t - 1] = [0, k + \lfloor D/q \rfloor]$.

Expanding via the product rule:

$$F^{(\ell)}(X) = \sum_{i=0}^{t-1} \sum_{\substack{\ell_0,\ell_1,\ldots,\ell_i \\ \sum \ell_j = \ell}} F_i^{(\ell_0)}(X) \cdot \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X).$$

A summand vanishes if $\ell_0 > k$, since $\deg(F_i) \leq k$ forces $F_i^{(\ell_0)} = 0$. It vanishes if some $\ell_j \notin \{0, 1, q\}$, since $\Lambda^{(\ell_j)} = 0$ for such ℓ_j. It also vanishes mod Λ if some $\ell_j = 0$, since that contributes a factor $\Lambda^{(0)} = \Lambda$. Thus, every nonzero summand must have $\ell_0 \in [0, k]$ and $\ell_1, \ldots, \ell_i \in \{1, q\}$.

When $\ell \bmod q \in [k+t, q-1]$, no such decomposition is possible: a sum of one integer from $[0,k]$ and at most $t-1$ integers from $\{1,q\}$ has residue mod q at most $k+(t-1) < k+t$. So every summand vanishes, giving $F^{(\ell)}(X) = 0$. ■

From this lemma we can get a relation of size of a set of zero and the degree of then interpolated pseudopolynomials which becomes zero at the set. For any $S \subseteq \mathbb{F}_q$, and any c, k, M satisfying

$$|S| < \frac{c}{M} \cdot k,$$

there exists a nonzero k-pseudopolynomial $F(X)$ of degree $D < cq$ such that for all $\alpha \in S$, $\mathrm{mult}(F,\alpha) \geq M$. This follows from dimension and constraint counting.

In the reverse direction, the following theorem shows that when q is prime, every nonzero k-pseudopolynomial of degree $D < cq$ has at most $2\frac{c}{M}k$ roots of multiplicity at least M.

Theorem 1.1.3 High Multiplicity Zeroes Bound, [Kop26]

Let q be prime, and let $F(X)$ be a nonzero k-pseudopolynomial of degree $D < cq$ over $\mathbb{F}_q$. Suppose M satisfies $c < M < q$. Then:

$$\left|\{\alpha \in \mathbb{F}_q \mid \mathrm{mult}(F,\alpha) \geq M\}\right| \leq \min\left(\frac{c}{M-c+1} \cdot k + c, \quad k\right).$$

We will not prove this theorem. The proof uses some arguments from [GK14] on wronskian ranks. The full proof of this theorem is in [Kop26, Section 5.7].

1.1.1.3 Codes From Pseudopolynomials

Now using Pseudopolynomials we can create codes similar to univariate multiplicity codes. Let $\Sigma = \mathbb{F}_q^M$. The codewords lie in $\Sigma^{\mathbb{F}_q}$: for each k-pseudopolynomial $F(X)$ of degree $D < cq$, the codeword $\mathrm{Enc}(F) : \mathbb{F}_q \to \Sigma^{\mathbb{F}_q}$ is defined:

$$\mathrm{Enc}(F)(\alpha) = \left(F^{(0)}(\alpha),\ F^{(1)}(\alpha),\ \ldots,\ F^{(M-1)}(\alpha)\right).$$

This code has cardinality $|\Sigma|^{ck/M}$, block length q, rate $R = \frac{c}{M} \cdot \frac{k}{q}$, and minimum distance at least

$$\left(1 - \min\left(\frac{k}{q},\ \frac{c}{M-c+1} \cdot \frac{k}{q} + \frac{c}{q},\ \frac{c}{M}\right)\right) \cdot q,$$

which is always at least $(1-2R) \cdot q$.

1.1.2 Twisted Pseudopolynomials

Definition 1.1.3: r-Twisted Pseudopolynomial

Suppose $r : \mathbb{F}_q \to \mathbb{F}_q$ is a function. Let $k < q$. We say $F(X) \in \mathbb{F}_q[X]$ is an r-twisted (h,M)-pseudopolynomial if for all $\ell < M$, there is some $U_\ell(X) \in \mathbb{F}_q[X]$ of degree at most h such that:

$$F_{\langle \ell \rangle} = r \cdot U_\ell.$$

The following lemma shows that any two r-twisted (h,M)-pseudopolynomials with degree at most cq are very closely related, provided h and c are small enough.

Lemma 1.1.4

Let c, h, M be parameters, with $c < M/2$. Let $r : \mathbb{F}_q \to \mathbb{F}_q$. Suppose $F(X), G(X) \in \mathbb{F}_q[X]$ with $\deg(F), \deg(G) < cq$

are both r-twisted (h, M)-pseudopolynomials and $k > \frac{M}{M-2c} \cdot (h+1)$. Then there exist nonzero k-pseudopolynomials $A(X), B(X)$, with $\deg(A), \deg(B) < Mq$, such that:

$$A(X) \cdot F(X) = B(X) \cdot G(X).$$

Proof: Now if $k \geq q$ then the result is immediate. So assume $k < q$. Since F, G both are r-twisted (h, M)-pseudopolynomial for each $\ell < M$, we have:

$$F_{\langle \ell \rangle} = r \cdot U_\ell, \qquad G_{\langle \ell \rangle} = r \cdot V_\ell,$$

where $U_\ell(X), V_\ell(X) \in \mathbb{F}_q[X]$ have degrees at most h.

The goal is to find A and B in the base-Λ form

$$A(X) = \sum_{i=0}^{M-c-1} A_i(X)\, \Lambda(X)^i, \qquad B(X) = \sum_{i=0}^{M-c-1} B_i(X)\, \Lambda(X)^i,$$

with $\deg(A_i), \deg(B_i) \leq k$, so that $AF = BG$. Treating the coefficients of the A_i and B_i as unknowns, their total count exceeds $2(M-c) \cdot k$.

The condition $AF = BG$ is captured by requiring $(A \cdot F)_{\langle \ell \rangle} = (B \cdot G)_{\langle \ell \rangle}$ for each $\ell < M$. Expanding each pseudo-derivative using the twisted conditions on F and G, these equalities can be packaged into a single identity in $\mathbb{F}_q[X, T]$:

$$\left(\sum_{\ell_1 < M} A_{\langle \ell_1 \rangle}(X) \cdot T^{\ell_1} \right) \cdot \left(\sum_{\ell_2 < M} U_{\ell_2}(X) \cdot T^{\ell_2} \right) = \left(\sum_{\ell_3 < M} B_{\langle \ell_3 \rangle}(X) \cdot T^{\ell_3} \right) \cdot \left(\sum_{\ell_4 < M} V_{\ell_4}(X) \cdot T^{\ell_4} \right) \pmod{T^M}.$$

Each of the M coefficient equalities (in $\mathbb{F}_q[X]$, of degree at most $h+k$) is a homogeneous linear condition on the unknowns, giving at most $(h+k+1) \cdot M$ constraints in total. Since by assumption

$$(h+k+1) \cdot M < 2k \cdot (M-c),$$

the number of constraints is strictly less than the number of unknowns, guaranteeing a nonzero solution.

It remains to verify that any such solution satisfies $AF = BG$ as a polynomial identity, not merely on $\mathbb{F}_q$. The construction ensures $(A \cdot F - B \cdot G)_{\langle \ell \rangle} = 0$ for every $\ell < M$, so every point of $\mathbb{F}_q$ is a root of $AF - BG$ of multiplicity at least M. But

$$\deg(A \cdot F - B \cdot G) < cq + (M-c)q = Mq,$$

so the polynomial $AF - BG$ has more roots (counting multiplicity) than its degree, forcing $AF - BG = 0$. ■

The next lemma shows that if we have many r-twisted (h, M)-pseudopolynomials with degree at most cq and c, h small, then they are nontrivially related. The smallness requirement on c is weaker, but the deduced relation is also weaker.

Lemma 1.1.5

Let m, c, h, M be parameters, with $c < \frac{m-1}{m} \cdot M$. Let $r : \mathbb{F}_q \to \mathbb{F}_q$. Suppose $F_1(X), \ldots, F_m(X) \in \mathbb{F}_q[X]$ with $\deg(F_i) < cq$ are all r-twisted (h, M)-pseudopolynomials. Let

$$k > \frac{M}{(m-1)M - mc} \cdot (h+1).$$

Then there exist k-pseudopolynomials $A_1(X), \ldots, A_m(X)$ with $\deg(A_i) < Mq$ and not all zero, such that, $\sum_i A_i(X)\, F_i(X) = 0$.

The proof is exactly like that of Lemma 1.1.4, which is the $m = 2$ case. The exact same argument also gives robust version.

Theorem 1.1.6 Robust Version for Two

Let m, c, h, γ, M be parameters, with $c < \frac{1}{2}(1-2\gamma)M$. Let $r_1, r_2 : \mathbb{F}_q \to \mathbb{F}_q$, with: $\Delta(r_1, r_2) \leq \gamma q$. Suppose $F(X), G(X) \in \mathbb{F}_q[X]$ with $\deg(F), \deg(G) < cq$ are r_1-twisted and r_2-twisted (h, M)-pseudopolynomials respectively. Let

$$k > \frac{M}{(1-2\gamma)M - 2c} \cdot (h+1).$$

Then there exist k-pseudopolynomials $A(X), B(X)$, with $\deg(A), \deg(B) < Mq$, not all zero, such that, $A(X) \cdot F(X) = B(X) \cdot G(X)$.

Theorem 1.1.7 Robust Version for Many

Let m, c, h, γ, M be parameters, with $c < \frac{1}{m}((m-1) - m\gamma)M$. Let $r_1, \ldots, r_m : \mathbb{F}_q \to \mathbb{F}_q$, with:

$$\left|\{\alpha \in \mathbb{F}_q \mid r_i(\alpha) \neq r_j(\alpha) \text{ for some } i, j\}\right| \leq \gamma q.$$

Suppose $F_1(X), \ldots, F_m(X) \in \mathbb{F}_q[X]$ with $\deg(F_i) < cq$ are r_i-twisted (h, M)-pseudopolynomials. Let

$$k > \frac{M}{(m-1-m\gamma)M - mc} \cdot (h+1).$$

Then there exist k-pseudopolynomials $A_1(X), \ldots, A_m(X)$, with $\deg(A_i) < Mq$, not all zero, such that, $\sum_i A_i(X)\, F_i(X) = 0$.

§ 1.2 Decoding Multiplicative Character Evaluations

Let $\psi \in \mathcal{X}_q$ is a non-trivial multiplicative character of $\mathbb{F}_q$ where $\mathrm{ord}(\psi) = d$. Then the polynomial based code we can construct is $(\psi \circ f(\alpha))_{\alpha \in \mathbb{F}_q}$ where $f \in \mathbb{F}_q[X]$. Now suppose g is another polynomial. Then for any $\alpha \in \mathbb{F}_q$,

$$\psi((f \cdot g^d)(\alpha)) = \psi(f(\alpha)) \cdot \psi^d(g(\alpha)) = \psi(f(\alpha))$$

So we assume f doesn't have a power d factor. Similarly, for any α, β where $\beta \neq 0$

$$\psi((\beta^d \cdot f)(\alpha)) = \psi^d(\beta) \cdot \psi(f(\alpha)) = \psi(f(\alpha))$$

Since different leading coefficient can lead to same character evaluation we assume f to be monic.

1.2.1 Quadratic Character

Now we take the quadratic character $\eta \in \mathcal{M}_q$. Hence, we assume the polynomials are monic and squarefree. Then by Theorem **??** we have

$$\left|\sum_{\alpha} \eta((g_1 \cdot g_2)(\alpha))\right| < O(d\sqrt{q})$$

Therefore the vectors $\eta \circ g_1$ and $\eta \circ g_2$ differs in at least $\left(q - O(d\sqrt{q})\right) = \left(\frac{1}{2} - \frac{d}{\sqrt{q}}\right) q$ many locations. So their hamming distance is at least $\left(\frac{1}{2} - \frac{d}{\sqrt{q}}\right) q$ which gives the distance of the code.

Below we give an algorithm to decode these codes from 1/4th of the minimum distance of the code. The decoding algorithm follows the *Berlekamp-Welch* of *Reed-Solomon codes* to some extent. Let g is the closest codeword. So suppose $G(X) := g^{(q-1)/2}(X)$.

If there had been zero error in the received word then we know the polynomial $\tilde{G}$ is $\psi \circ g$-twisted $(O(dM), M))$-pseudopolynomial. But we have the received word very close to the original codeword $\psi \circ g$. So we will try to find a

r-twisted pseudopolynomial F with good parameters such that we can use Lemma 1.1.4 or Theorem **??** to relate them. So first we have to show such F exists.

1.2.1.1 Constructing r-twisted Pseudopolynomial

Let $S = \{\alpha \in \mathbb{F}_q \colon \psi \circ g(\alpha) \neq r(\alpha)\}$ be the error locator set. Then consider the error locator polynomial $Z_S(X) = \prod_{\alpha \in S}(X - \alpha)^M$. Then all elements of S are roots of Z_S with multiplicity at M. So we take any multiple of Z_S, $E(X)$ which is a h-pseudopolynomial of degree $< c$ where h, c, M are parameters to be chosen later. So by construction $E(X) = \sum_{i=0}^{c-1} E_i(X) \cdot \Lambda^i(X)$ by Lemma 1.1.1. So $\deg(E_i) \leq h$. Therefore the number of coefficients is $c(h+1)$. But number of constraints is $e \cdot M$. Hence we want $c(h+1) > e \cdot M$. So it is enough to have $ch = eM$.

So now we have a nonzero multiple of $Z_S(X)$ such that for all $\alpha \in S$, $\mathrm{mult}(E, \alpha) \geq M$. Therefore we take the polynomial $G(X) \cdot E(X)$ where $G(X) = g^{\frac{q-1}{m}+M}(X)$. We will show that this F is a r-twisted $(h + dM, M)$-pseudopolynomial.

First we will calculate $E_{\langle \ell \rangle}$ for all $0 \leq \ell < M$. By Lemma A.4.2 we have

$$E^{(\ell)}(X) = \sum_{i=0}^{c-1} (-1)^i E_i^{(\ell-i)}(X) \bmod \Lambda(X) \implies E_{\langle \ell \rangle}(X) = \sum_{i=0}^{c-1} (-1)^i E_i^{(\ell-i)}(X)$$

Therefore, for any $\alpha \in \mathbb{F}_q$ we have

$$\begin{aligned}
(E \cdot G)^{(\ell)}(\alpha) &= \sum_{\ell_1+\ell_2=\ell} E^{(\ell_1)}(\alpha) \cdot G^{(\ell_2)}(\alpha) \\
&= \sum_{\ell_1+\ell_2=\ell} E_{\langle \ell_1 \rangle}(\alpha) \cdot G_{g,w,\ell_2}(\alpha) \cdot g^{M-\ell_2}(\alpha) \cdot (\psi \circ g(\alpha)) && \text{[By Theorem A.4.4]} \\
&= \psi \circ g(\alpha) \sum_{\ell_1+\ell_2=\ell} E_{\langle \ell_1 \rangle}(\alpha) \cdot G_{g,w,\ell_2}(\alpha) \cdot g^{M-\ell_2}(\alpha)
\end{aligned}$$

So define

$$V_\ell(X) = \sum_{\ell_1+\ell_2=\ell} E_{\langle \ell_1 \rangle}(X) \cdot G_{g,w,\ell_2}(X) \cdot g^{M-\ell_2}(X).$$

Then we have $(E \cdot G)^{(\ell)}(\alpha) = r(\alpha) \cdot V_\ell(\alpha)$. Since at error locations we have $E^{(i)}(\alpha)$ for all $0 \leq i < M$ we have $V_\ell(\alpha) = 0$. So this equality holds. Now $\deg(V_\ell) \leq d\left(\frac{q-1}{2} + M\right) cq =: D$ and $\deg(V_\ell) \leq h + dM$. So we have F to be a r-twisted $(h + dM, M)$-pseudopolynomial of degree D. So if we find any r-twisted $(h + dM, M)$-pseudopolynomial F of degree D via interpolation then the $E \cdot G$ and F, are both r-twisted. So by Lemma 1.1.4 there exists $A, B \in \mathbb{F}_q[X]$ such that $A \cdot (EG) = B \cdot F$. We can rewrite and say there exists $A, B \in \mathbb{F}_q[X]$ such that $A \cdot G = B \cdot F(X)$. So we have the following algorithm:

Algorithm 1: Decoding from $1/8$–th of distance of Quadratic Character Evaluation

Input: Noisy received word, $r : \mathbb{F}_q \to \{0, 1, -1\}$, with degree $d \leq O(\epsilon\sqrt{q})$ and error bound $e \leq (1/8 - \epsilon)\, q$

Output: Find the closest vector $\eta \circ f$ where $f \in \mathbb{F}_q[X]$ of degree at most d

```
begin
    Set M = 16/ϵ · d, c = M/2, h = 2 · e, D = d((q−1)/2 + M) + cq
    Use system of linear equations to find r-twisted (h + dM, M)-pseudopolynomial F(X) of degree at most D.
    Complete factorization of F(X) into distinct monic irreducible factors {h_1, ..., h_k} with mult(h_j, F) = k_j.
    Compute the set J = {j ∈ [t] : k_j ∈ [3q/8, 7q/8] mod q}. Then compute f(X) = ∏_{j∈J} h_j.
    return f(X)
```

1.2.1.2 Relating F, G and Factor Multiplicities

To make sense of the last step of separating out the high multiplicity factors we first have to look into A, B.

Step 1: Relating F and G We now show that any polynomial $F(X)$ obtained in the algorithm must satisfy a nontrivial algebraic relation with $G(X)$. Let $U_0, \ldots, U_{M-1}$ be the polynomials such that $F^{(\ell)}(\alpha) = r(\alpha) \cdot U_\ell(\alpha)$ for all $0 \le \ell < M$, $\alpha \in \mathbb{F}_q$. So now we set parameters $t = 3/8M$ and $k = e + 4dM$. We claim that there exist nonzero polynomials $A(X)$ and $B(X)$ of the form

$$A(X) = \sum_{i=0}^{t-1} A_i(X)\Lambda^i(X), \qquad B(X) = \sum_{i=0}^{t+c-1} B_i(X)\Lambda^i(X),$$

with $\deg(A_i) \le k$, $\deg(B_i) \le k + 2e$, such that $A(X)F(X) = B(X)G(X)$. We again argue by dimension counting. The total number of coefficients of the A_i and B_i equals $N = t(k+1) + (t+c)(k+2e+1)$. Substituting the values of t and k, we get

$$N > \frac{3}{8}M(e + 4dM) + \frac{7}{8}M(3e + 4dM) = 3Me + 5dM^2. \tag{1.1}$$

We regard these coefficients as variables and impose homogeneous linear conditions so that for every $0 \le \ell < M$ and every $\alpha \in \mathbb{F}_q \setminus S$,

$$(A \cdot F)^{(\ell)}(\alpha) = (B \cdot G)^{(\ell)}(\alpha). \tag{1.2}$$

Consequently, the polynomial $A(X)F(X) - B(X)G(X)$ vanishes to multiplicity at least M at every point of $\mathbb{F}_q \setminus S$. Hence it has at least

$$M \cdot |\mathbb{F}_q \setminus S| \ge M(q - e) > \left(\frac{7}{8} + \varepsilon\right) Mq$$

roots counted with multiplicity. On the other hand,

$$\deg(A \cdot F - B \cdot G) \le \max\{\deg(A \cdot F), \deg(B \cdot G)\} < \left(\frac{7}{8} + \frac{\varepsilon}{2}\right) Mq.$$

Thus, $A(X)F(X) = B(X)G(X)$. It remains to show that Equation (1.2) can indeed be enforced using fewer than N linear constraints. For each ℓ, define $A_{\langle \ell \rangle}(X) = \sum_{i=0}^{t-1} (-1)^i A_i^{(\ell-i)}(X)$, $B_{\langle \ell \rangle}(X) = \sum_{i=0}^{t+c-1} (-1)^i B_i^{(\ell-i)}(X)$. By Lemma A.4.2 we get $A^{(\ell)}(\alpha) = A^{\langle \ell \rangle}(\alpha)$ and $B^{(\ell)}(\alpha) = B^{\langle \ell \rangle}(\alpha)$ for every $\alpha \in \mathbb{F}_q$. Now recall that $F^{(\ell)}(\alpha) = r(\alpha)U_\ell(\alpha)$. Hence,

$$(A \cdot F)^{(\ell)}(\alpha) = \sum_{\ell_1 + \ell_2 = \ell} A^{(\ell_1)}(\alpha) F^{(\ell_2)}(\alpha) \implies (A \cdot F)^{(\ell)}(\alpha) = r(\alpha) \cdot \sum_{\ell_1 + \ell_2 = \ell} A_{\langle \ell_1 \rangle}(\alpha) U_{\ell_2}(\alpha). \tag{1.3}$$

Similarly, we get

$$G^{(\ell)}(\alpha) = (\chi \circ g)(\alpha)\, H_{g,w,\ell}(\alpha)\, g^{M-\ell}(\alpha), \implies (B \cdot G)^{(\ell)}(\alpha) = (\chi \circ g)(\alpha) \cdot \sum_{\ell_1 + \ell_2 = \ell} B_{\langle \ell_1 \rangle}(\alpha) G_{g,w,\ell_2}(\alpha) g^{M-\ell_2}(\alpha). \tag{1.4}$$

We therefore impose, for each $0 \le \ell < M$, the polynomial identity

$$\sum_{\ell_1 + \ell_2 = \ell} A_{\langle \ell_1 \rangle}(X) U_{\ell_2}(X) = \sum_{\ell_1 + \ell_2 = \ell} B_{\langle \ell_1 \rangle}(X) G_{g,w,\ell_2}(X) g^{M-\ell_2}(X). \tag{1.5}$$

Both sides have degree at most $3e + dM$. Hence, the total number of linear conditions arising from all these identities is at most $M(3e + dM + 1)$, which is strictly smaller than N by Equation (1.1). Therefore there exists a nontrivial choice of $A(X), B(X)$ satisfying Equation (1.5). Finally, combining Equations (1.3), (1.4), and (1.5), and using the fact that $r(\alpha) = (\chi \circ g)(\alpha)$ for all $\alpha \in \mathbb{F}_q \setminus S$, we obtain Equation (1.2), and hence $A(X)F(X) = B(X)G(X)$.

Step 2: Recovering the factors of $g(X)$ We now use the identity $A(X)F(X) = B(X)G(X)$ to compare the irreducible factors of $F(X)$ and $G(X)$. The essential point is that the special form of $A(X)$ and $B(X)$ forces all their irreducible factors to occur with multiplicity very close to a multiple of q by Lemma 1.1.2. In contrast, the factors arising from $g(X)$ occur inside $G(X)$ with multiplicity approximately $q/2$ modulo q. Since $g(X)$ is squarefree, this allows us to identify precisely the irreducible factors of $g(X)$.

Now let $H(X)$ be an irreducible polynomial in $\mathbb{F}_q[X]$. Denote by $\mu_F, \mu_A, \mu_B, \mu_G, \mu_g$ the multiplicities of $H(X)$ in $F(X), A(X), B(X), G(X), g(X)$ respectively. Since $g(X)$ is squarefree, $\mu_g \in \{0, 1\}$. From the identity $AF = BG$, we obtain $\mu_F = \mu_G + \mu_B - \mu_A$. Applying the lemma to A and B, we have

$$\mu_B \in q\mathbb{Z} + [0, 3e + 4dM + \tfrac{7}{8}M] \quad \text{and} \quad \mu_A \in q\mathbb{Z} + [0, e + 4dM + \tfrac{3}{8}M].$$

Therefore $\mu_F \in \mu_G + q\mathbb{Z} + [-e - 5dM,\ 3e + 5dM]$. Now $\mu_G = \left(\frac{q-1}{2} + M\right)\mu_g$.

If $\mu_g = 1$, then $\mu_G = \frac{q}{2} + \frac{M-1}{2}$, and hence $\mu_F \in q\mathbb{Z} + \frac{q}{2} + [-e - 5dM,\ 3e + 6dM]$. Using $e \leq \left(\frac{1}{8} - \varepsilon\right) q$ and $6dM < \varepsilon q$, we obtain $\mu_F \in q\mathbb{Z} + (3q/8, 7q/8)$. On the other hand, if $\mu_g = 0$, then $\mu_G = 0$, and thus

$$\mu_F \in q\mathbb{Z} + [-e - 5dM,\ 3e + 5dM] \subseteq q\mathbb{Z} + (-q/8, 3q/8).$$

Hence, the set J selected in Step 5 of the algorithm consists precisely of those irreducible factors which divide $g(X)$. Therefore, the polynomial produced in Step 5 is exactly $f(X) = g(X)$.

1.2.2 General Multiplicative Character of order m

Now we take the multiplicative character $\psi \in \boldsymbol{\mathcal{M}}_q$or order $m > 1$. Hence, we assume the polynomials are monic and don't have a factor of power m. Then by Theorem **??** we have $|\sum_\alpha \psi((g_1 \cdot g_2)(\alpha))| < O(m\sqrt{q})$. Therefore the hamming distance is at least $\left(1 - \frac{1}{m}\right) q$ which gives the distance of the code. So the unique decoding radius is $\frac{1}{2}\left(1 - \frac{1}{m}\right) q$.

Now we can also think of the character as $\psi : \mathbb{F}_q \to \{0\} \cup \mathbb{F}_q^{*((q-1)/m)}$ where $\alpha \mapsto \alpha^{(q-1)/m}$. So here we will consider the multiplicative as an map from $\mathbb{F}_q \to \{0\} \cup \mathbb{F}_q^{*((q-1)/m)}$. Like for the case of quadratic character we will not prove the correctness of the algorithm instead we just give the pseudocode of the algorithm below. The details of the correctness of the algorithm is given in [Kop26].

Algorithm 2: Decoding from $1/12$–th of distance of mth-order Character Evaluation

Input: Noisy received word, $r : \mathbb{F}_q \to \{0, 1, -1\}$, with degree $d \leq O(\frac{\epsilon}{m}\sqrt{q})$ and error bound $e \leq (1/12 - \epsilon)\, q$

Output: Find the closest vector $\psi \circ f$ where $f \in \mathbb{F}_q[X]$ of degree at most d

1 **begin**

2 Set $M = 16/\epsilon \cdot dm$, $c = M/2$, $h = 2 \cdot e$, $D = d((q-1) \cdot (m-1)/m + M) + cq$

3 **for** $k \in [m-1]$ **do**

4 Use system of linear equations to find r^k-twisted $(h + (m-1)dM, M)$-pseudopolynomial $F_k(X)$ of degree at most D.

5 Complete factorization of $F(X)$ into distinct monic irreducible factors $\{h_{1,k}, \ldots, h_{t_k,k}\}$ with $\mathrm{mult}(h_{j,k}, F_k) = d_{j,k}$.

6 Let the irreducible factors are $H_1, \ldots, H_t$ with $\mathrm{mult}(H_j, F_k) = d_{j,k}$.

7 **for** $j \in [t]$ **do**

8 Let $\mu_j \in \{0, 1, \ldots, m-1\}$ be the unique number such the for all $k \in [m-1]$

$$d_{k,j} - \frac{k \cdot \mu_j}{m} \cdot (q-1) \in \left(-\left(\frac{1}{12} - \epsilon\right) q, \left(\frac{3}{12} - \epsilon\right) q\right) \bmod q$$

9 Compute $f(X) = \prod_{j=1}^{t} h_j^{\mu_j}(X)$.

10 **return** $f(X)$

§ 1.3 Decoding Dual BCH Codes (Additive Character Evaluations)

In this section we basically decode the famous Dual BCH codes. We first discuss the code construction. Let q be a power of 2, say $q = 2^b$. Let Tr be the absolute trace function from $\mathbb{F}_q \to \mathbb{F}_2$. Let $\mathrm{Tr}(X)$ be the polynomial $\mathrm{Tr}(X) = \sum_{i=0}^{b-1} X^{2^i}$. Now suppose $\deg(g) = d$. Then we can write g as

$$g(X) = a + \tilde{g}(X) + h(X) + h^2(X)$$

where $a \in \mathbb{F}_q$, $\tilde{g}, h \in \mathbb{F}_q[X]$ which satisfies the following:

(i) $\tilde{g}$ has only odd degree monomials

(ii) $h(X)$ has 0 constant term

Now for any $\alpha \in \mathbb{F}_q$, we have $\mathrm{Tr} \circ g(\alpha) = \mathrm{Tr}(a) + \mathrm{Tr} \circ \tilde{g}(\alpha)$. Therefore, either $\mathrm{Tr} \circ g = \mathrm{Tr} \circ \tilde{g}$ or $\mathrm{Tr} \circ g = 1 + \mathrm{Tr} \circ \tilde{g}$. So we assume g only has odd degree monomials. So now take the $\chi_1 \in \boldsymbol{\chi}_q$ canonical additive character of $\mathbb{F}_q$. Then for any α, $\chi_1(\alpha) = (-1)^{\mathrm{Tr}(\alpha)}$. So for a polynomial $g \in \mathbb{F}_q[X]$ of degree at most d, the encoding of the dual BCH code is

$$\mathrm{Enc}(g) = \left(\chi_1 \circ g(\alpha)\right)_{\alpha \in \mathbb{F}_q}$$

by Theorem **??** we have

$$\left| \sum_{\alpha \in \mathbb{F}_q} \chi_1 \circ g(\alpha) \right| \leq (d-1)\sqrt{q}$$

Therefore for any two distinct polynomials $g_1, g_2 \in \mathbb{F}_q[X]$ of degree d we have

$$\left| \sum_{\alpha \in \mathbb{F}_q} \chi_1 \circ (g_1 - g_2)(\alpha) \right| \leq O(d\sqrt{q})$$

So if $d = o(\sqrt{q})$ then we have $\Delta(\mathrm{Tr} \circ g_1, \mathrm{Tr} \circ g_2) \in \left(\frac{1}{2} \pm O\left(\frac{d}{\sqrt{q}}\right)\right) q$.

Chapter 2

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Chapter A

Appendix

§ A.1 Continued Fractions over Finite Fields

Let $r_0, r_1 \in \mathbb{F}_q[X]$ with $r_1 \neq 0$. Applying the Euclidean algorithm yields a sequence

$$r_i = A_i r_{i+1} + r_{i+2}, \quad i = 0, 1, \ldots, s,$$

where $0 \leq \deg(r_{i+1}) < \deg(r_i)$ for $i = 1, \ldots, s$ and $r_{s+2} = 0$. Here $A_0, A_1, \ldots, A_s \in \mathbb{F}_q[X]$, with $A_1, \ldots, A_s$ of positive degree. This gives the *continued fraction expansion*

$$\frac{r_0}{r_1} = A_0 + \cfrac{1}{A_1 + \cfrac{1}{A_2 + \cdots}} =: [A_0, A_1, A_2, \ldots, A_s].$$

Define polynomials $P_i, Q_i \in \mathbb{F}_q[X]$ for $i = -1, 0, 1, \ldots, s$ by the recurrences

$$\begin{aligned} P_{-1} &= 1, \quad P_0 = A_0, \quad P_i = A_i P_{i-1} + P_{i-2} \quad (i = 1, \ldots, s), \\ Q_{-1} &= 0, \quad Q_0 = 1, \quad Q_i = A_i Q_{i-1} + Q_{i-2} \quad (i = 1, \ldots, s). \end{aligned}$$

It is clear that $\deg(P_{i-1}) < \deg(P_i)$ and $\deg(Q_{i-1}) < \deg(Q_i)$ for $i = 1, \ldots, s$. We extend the definition of degree by $\deg(f/g) = \deg(f) - \deg(g)$ for a rational function f/g.

Lemma A.1.1 Rational Approximants via Convergents

For any rational function ρ of nonnegative degree,

$$[A_0, A_1, \ldots, A_{i-1}, \rho] = \frac{\rho P_{i-1} + P_{i-2}}{\rho Q_{i-1} + Q_{i-2}} \quad \textit{for } i = 1, \ldots, s+1.$$

Proof: We proceed by induction on i. For $i = 1$ both sides equal $A_0 + \rho^{-1} = \rho^{-1}(A_0\rho + 1) = (\rho P_0 + P_{-1})/(\rho Q_0 + Q_{-1})$. Suppose the identity holds for some i, $1 \leq i \leq s$. Since $A_i + \rho^{-1}$ has positive degree, applying the inductive hypothesis gives

$$[A_0, \ldots, A_i, \rho] = \frac{(A_i + \rho^{-1})P_{i-1} + P_{i-2}}{(A_i + \rho^{-1})Q_{i-1} + Q_{i-2}} = \frac{\rho P_i + P_{i-1}}{\rho Q_i + Q_{i-1}},$$

using the recurrence relations. ■ ■

Corollary A.1.2 Convergents Formula

For $i = 0, 1, \ldots, s$ we have $[A_0, A_1, \ldots, A_i] = P_i/Q_i$.

Lemma A.1.3 Representation of r_0/r_1 via Convergents

For $i = 0, 1, \ldots, s$ we have

$$\frac{r_0}{r_1} = \frac{P_i + \beta_i P_{i-1}}{Q_i + \beta_i Q_{i-1}},$$

where $\beta_i = r_{i+2}/r_{i+1}$ is a rational function of negative degree.

Proof: We proceed by induction. For $i = 0$, $(P_0 + \beta_0 P_{-1})/(Q_0 + \beta_0 Q_{-1}) = (A_0 + r_2/r_1)/(1) = (A_0 r_1 + r_2)/r_1 = r_0/r_1$. The inductive step follows from the recurrences and $\beta_i^{-1} = A_{i+1} + \beta_{i+1}$. ■ ■

Lemma A.1.4 Determinant Identity for Convergents

For $i = 0, 1, \ldots, s$ we have $P_i Q_{i-1} - P_{i-1} Q_i = (-1)^{i-1}$.

Proof: Induction. For $i = 0$: $P_0 Q_{-1} - P_{-1} Q_0 = 0 \cdot 0 - 1 \cdot 1 = -1 = (-1)^{-1}$. Assuming the identity at i,

$$P_{i+1} Q_i - P_i Q_{i+1} = (A_{i+1} P_i + P_{i-1}) Q_i - P_i (A_{i+1} Q_i + Q_{i-1}) = -(P_i Q_{i-1} - P_{i-1} Q_i) = (-1)^i. \blacksquare$$

■

Corollary A.1.5 Coprimality of Convergents

For $i = 0, 1, \ldots, s$ we have $\gcd(P_i, Q_i) = 1$.

Proof: If $d_i = \gcd(P_i, Q_i)$ then Lemma A.1.4 shows d_i divides $(-1)^{i-1}$, so $d_i = 1$. ■ ■

§ A.2 A Product Identity for Roots of Unity

The following identity, used in the factorisation of lifted L-functions, expresses a product of linear factors twisted by roots of unity as a perfect power.

Lemma A.2.1 Roots-of-Unity Product Identity

Let $n, m \in \mathbb{N}$ and set $d = \gcd(m, n)$. Then

$$\prod_{t=1}^{n} \left(1 - e^{2\pi i m t/n} X\right) = \left(1 - X^{n/d}\right)^d.$$

Proof: Let $\omega = e^{2\pi i/n}$ be a primitive n-th root of unity, so the factors are $1 - \omega^{mt} X$ for $t = 1, \ldots, n$. Since $t = n$ gives $\omega^{mn} = 1 = \omega^0$, the product is the same as $\prod_{t=0}^{n-1} (1 - \omega^{mt} X)$.

Because $d = \gcd(m, n)$, the element ω^m is a primitive (n/d)-th root of unity. As t ranges over $\{0, 1, \ldots, n-1\}$, the values ω^{mt} cycle through the (n/d)-th roots of unity exactly d times each. Hence

$$\prod_{t=0}^{n-1} \left(1 - \omega^{mt} X\right) = \left(\prod_{k=0}^{n/d-1} \left(1 - (\omega^m)^k X\right)\right)^d.$$

Since ω^m is a primitive (n/d)-th root of unity, the inner product equals $1 - X^{n/d}$ by the standard factorisation of $1 - Y^N = \prod_{k=0}^{N-1}(1 - \zeta^k Y)$. Therefore the whole product equals $(1 - X^{n/d})^d$. ■ ■

§ A.3 Elementary Symmetric Polynomials

Waring's Formula expresses each power-sum symmetric polynomial $s_k = x_1^k + x_2^k + \cdots + x_n^k$ as an explicit polynomial in the elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$. It is the source of the closed-form reduction of lifted Kloosterman sums $K(\chi^{(s)}; a, b)$ to the ground-field sum $K(\chi; a, b)$ in Lemma **??**: applying Waring's formula to $\omega_1^s + \omega_2^s$ with $\sigma_1 = \omega_1 + \omega_2 = -K(\chi; a, b)$ and $\sigma_2 = \omega_1\omega_2 = q$ produces the binomial-coefficient expansion. We state the general identity here for reference.

Theorem A.3.1 Waring's Formula

Let $x_1, \ldots, x_n$ be indeterminates, let

$$\sigma_j = \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} x_{i_1} x_{i_2} \cdots x_{i_j} \qquad (j = 1, \ldots, n)$$

be the elementary symmetric polynomials in $x_1, \ldots, x_n$, and let

$$s_k = x_1^k + x_2^k + \cdots + x_n^k$$

be the kth power-sum. Then for every integer $k \ge 1$,

$$s_k = \sum (-1)^{i_2 + i_4 + i_6 + \cdots} \cdot \frac{(i_1 + i_2 + \cdots + i_n - 1)! \cdot k}{i_1! \cdot i_2! \cdots i_n!} \cdot \sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n},$$

where the summation runs over all n-tuples $(i_1, \ldots, i_n)$ of non-negative integers satisfying $i_1 + 2i_2 + \cdots + ni_n = k$. Each coefficient of $\sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n}$ is an integer.

Theorem A.3.2

Suppose $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ is a symmetric polynomial. Let $\overline{X} = (X_1, \ldots, X_n)$. Then there exists a polynomial $g \in \mathbb{F}_q[Y_1, \ldots, Y_n]$ such that

$$f(X_1, \ldots, X_n) = g(\mathrm{ESym}_1(\overline{X}), \ldots, \mathrm{ESym}_n(\overline{X})).$$

Moreover,

(i) if $\deg_{X_i}(f) = d$ for all $i \in [n]$ then $\deg G = d$.

(ii) if $\deg f = D$ then for every monomial $Y_1^{d_1} \cdots Y_n^{d_n}$ of g with non-zero coefficient then $d_1 + 2 \cdot d_2 + \cdots + n \cdot d_n = D$.

§ A.4 Hasse Derivative

Definition A.4.1: Hasse Derivative

Let $\mathbb{F}$ be any field. Let $P \in \mathbb{F}[X_1, \ldots, X_n]$. Then $\bar{i} \in \mathbb{Z}_0^n$ Hasse derivative of P is denoted by $P^{(\bar{i})}(X_1, \ldots, X_n)$ or $H^{(\bar{i})}(P)$ which is the coefficient of $Z^{i_1} \cdots Z_n^{i_n} =: \overline{Z}^{\bar{i}}$ in the polynomial $P(X_1 + Z_1, \ldots, X_n + Z_n)$. Thus

$$P(X_1 + Z_1, \ldots, X_n + Z_n) = \sum_{\bar{i}} P^{(\bar{i})}(X_1, \ldots, X_n) \cdot \overline{Z}^{\bar{i}}$$

For univariate polynomials we immediately get a closed form formula of kth hasse derivative. Let $f \in \mathbb{F}[X]$ where $f(X) = \sum_{i=0}^{d} a_i X^i$. Then

$$f^{(k)}(X) = \sum_{i=0}^{d} \binom{k}{i} \cdot a_i X^{i-k}$$

From the definition we have the following observation

Observation A.1. *For any $f, f_1, f_2 \in \mathbb{F}_q[X_1, \ldots, X_n]$ and $\alpha \in \mathbb{F}$ and for any $\bar{k} \in \mathbb{Z}^n$ we have*

(i) $(f_1 + f_2)^{(\bar{k})}(X_1, \ldots, X_n) = f_1^{(\bar{k})}(X_1, \ldots, X_n) + f_2^{(\bar{k})}(X_1, \ldots, X_n)$

(ii) $(\alpha \cdot f)^{(\bar{k})}(X_1, \ldots, X_n) = \alpha \cdot f^{(\bar{k})}(X_1, \ldots, X_n)$.

Since throughout this report we only use hasse derivatives in the context of univariate polynomials we will only focus on univariate polynomials and see some properties of hasse derivatives. Here we take $\mathbb{F}$ to be any field.

Lemma A.4.1

(i) For $f_1, \ldots, f_t \in \mathbb{F}[X]$ then we have

$$(f_1 \cdot f_t)^{(n)}(X) = \sum_{\substack{n_1, \ldots, n_t \geq 0 \\ n_1 + \cdots n_t = n}} \prod_{i=1}^{t} f_i^{(n_i)}(X)$$

(ii) For any $\alpha \in \mathbb{F}$ we have

$$\left((X - c)^t\right)^{(n)} = \binom{t}{n} \cdot (X - c)^{t-n}.$$

(iii) For $0 \leq n \leq$ and $f, g \in \mathbb{F}[X]$ we have

$$\left(gf^t\right)^{(n)}(X) = \tilde{g} \cdot f^{t-n}(X)$$

where $\tilde{g} \in \mathbb{F}[X]$ with $\deg(\tilde{g}) \leq \deg(g) + n(\deg(f) - 1)$.

Proof:

(i) Since the hasse derivative operator is linear it is enough to show this property for monomials. So suppose $f_j(X) = X^{k_j}$ for all $j \in [t]$. Then $f_1 \cdots f_t = X^{\sum_{j=1}^{t} k_j}$. Therefore,

$$(f_1 \cdots f_t)^{(n)}(X) = \binom{\sum_{j=1}^{t} k_j}{t}, \qquad \prod_{j=1}^{t} f_j^{n_j}(X) = \prod_{j=1}^{t} \binom{k_j}{n_j} \text{ for all } n_j \in \mathbb{Z}_0, j \in [t]$$

So it is enough to show that

$$\binom{\sum_{j=1}^{t} k_j}{t} = \sum_{\substack{n_1, \ldots, n_t \geq 0 \\ n_1 + \cdots n_t = n}} \prod_{j=1}^{t} \binom{k_j}{n_j}$$

But this is true by comparing the coefficients of X^n in $(X+1)^{\sum_{j=1}^t k_j} = \prod_{j=1}^t (X+1)^{k_j}$. So we have the result.

(ii) Now take $f_j(X) = X - c$ for all $j \in [t]$. Then $f_j^{(n)}(X) = 1$ if $n = 1$ and 0 if $n > 1$. Therefore, if we use the first part on $f_1, \ldots, f_j$ in the sum on right-hand side the only terms which survives are when each n_j are either 0 or 1. The number of such terms are $\binom{t}{n}$ and each term is $(X-c)^{t-n}$.

(iii) Again by using the first part we get

$$(gf^t)^{(n)}(X) = \sum_{\substack{n_0, n_1, \ldots, n_t \geq 0 \\ n_0 + n_1 + \cdots + n_t = n}} g^{(n_0)}(X) \prod_{j=1}^{t} f^{(n_j)}(X)$$

Now each summand is divisible by f^{t-n}. Hence, there exists some $\tilde{g} \in \mathbb{F}[X]$ such that $(gf^t)^{(n)}(X) = \tilde{g}(X) \cdot f^{t-n}(X)$. Furthermore,

$$\deg(\tilde{g}) = \deg\left((gf^t)^{(n)}\right) - (t-n)\deg(f) \leq \deg(g) + t\deg(f) - n - (t-n)\deg(f) = \deg(g) + (\deg(f) - 1)$$

So we have the lemma. ■

Now for our use-case in chapter 1 we use the polynomial $\Lambda(X) = X^q - X$ and its powers. And we compute its derivatives to do polynomial arithmetic.

Observation A.2. *For $\Lambda(X)$ and for any $\ell \in \mathbb{Z}_0$*

$$\Lambda^{(\ell)}(X) = \begin{cases} \Lambda(X) & \ell = 0 \\ -1 & \ell = 1 \\ 1 & \ell = q \\ 0 & \textit{otherwise} \end{cases}$$

Lemma A.4.2

Suppose $U(X), V(X) \in \mathbb{F}_q[X]$, $\deg(V) < q$, $i \geq 0$, and $U(X) = V(X) \cdot \Lambda^i(X)$. Then for $0 \leq \ell < q$,

$$U^{(\ell)}(X) \equiv (-1)^i \cdot V^{(\ell - i)}(X) \pmod{\Lambda(X)}.$$

Proof: Now using the Lemma A.4.1(i) we get

$$U^{(\ell)}(X) = \sum_{\substack{\ell_0, \ldots, \ell_i \geq 0 \\ \ell_0 + \cdots + \ell_i = \ell}} V^{(\ell_0)}(X) \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X)$$

Now if $\ell < i$ then in every summand one of $\ell_1, \ldots, \ell_i$ must equal 0. Therefore, all the terms are multiples of $\Lambda(X)$. So $U^{(\ell)}(X)$ is a multiple of $\Lambda(X)$.

Now from the above observation if $\ell \geq i$ then all but at most one of these terms is a multiple of $\Lambda(X)$. Since $\ell < q$ every term has all the $\ell_j < q$. Therefore, $\ell_1, \ldots, \ell_i \in \{0, 1\}$. If at least one of $\ell_1, \ldots, \ell_i$ equals to 0 then $\Lambda(X)$ survives as a factor of the term. So the only remaining term is $\ell_1 = \cdots = \ell_i = 1$. In this case $\ell_0 = \ell - i$ and therefore this term equals $V^{(\ell - i)}(X) \cdot (-1)^i$. Hence we have the lemma. ■

As a corollary of the Lemma A.4.1(i,iii) and the above lemma we get the following result which is very usefull to us in chapter 1.

Corollary A.4.3

Let $f \in \mathbb{F}_q[X]$ be a polynomial with $\deg(g) \leq d$. Then the ℓth derivative of $g^j(X)$ is of the form

$$(g^j)^{(\ell)}(X) = G_{g,j,\ell}(X) \cdot g(X)^{j-\ell}$$

where $G_{g,j,\ell}(X) \in \mathbb{F}_q[X]$ and $\deg(G_{g,j,\ell}) \leq d\ell - \ell$.

Theorem A.4.4

Suppose $\mathrm{char}(\mathbb{F}) = p$ where p is a prime. Let $Q(X,Y) \in \mathbb{F}[X,Y]$ be a polynomial and define $h(X) = Q\left(X, X^{p^s}\right)$ for some $s \in \mathbb{N}$. Then for all $0 \leq n < p^s$

$$h^{(n)}(X) = Q^{(n,0)}\left(X, X^{p^s}\right)$$

where $Q^{(i,0)}$ is the ith partial derivative of Q with respect to X where we consider Q as a polynomial of $\mathbb{F}[Y][X]$.

Proof: Since hasse derivative is a linear operator it suffices to prove the above for monomials of the form X^iY^j. So suppose $Q(X,Y) = X^iY^j$. Then $h(X) = Q(X, X^{p^s}) = X^i \cdot \left(X^{p^s}\right)^j$. Now take $f_1(X) = X^i$ and $f_2 = \cdots = f_{k+1} = X^{p^s}$. Then $h = f_1 \cdots f_{k+1}$. So by Lemma A.4.1(i)

$$h^{(n)}(X) = \sum_{\substack{n_1,\ldots,n_{k+1} \geq 0 \\ n_1 + \cdots n_{k+1} = n}} \prod_{i=1}^{k+1} f_i^{(n_i)}(X)$$

Since $p \mid \binom{p^s}{n}$ for all $0 \leq n < p^s$ we have

$$(X^{p^s})^{(n)} = \binom{p^s}{X}^{p^s - n} \equiv 0$$

Therefore the only term which survives is when $n_1 = \cdots n_{k+1} = 0$ and $n_0 = n$. That means $h^{(n)} = Q^{(n,0)}(X, X^{p^s})$. ■

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